

AYUSH GIRI

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Education

SRM University AP, Amaravati, India
B.Tech in Computer Science & Engineering

2023 – 2027
CGPA: 8.35/10

Technical Skills

Core: Python, Java, C++, JavaScript, TypeScript, SQL

Full-Stack: React, Next.js, FastAPI, Node.js, Express.js, RESTful API Design, JWT Authentication

Databases: PostgreSQL, MongoDB, SQLAlchemy (ORM)

Infra & Tooling: Docker, Redis, Celery, Socket.IO, GitHub Actions (CI), pytest

AI/LLM: Google Gemini API, Prompt Engineering, Schema-Validated LLM Pipelines

Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Research Experience

Research Intern

June 2025 - Aug 2025

SRM University AP

- Built distributed computation workflows and feature extraction frameworks to improve machine learning experimentation efficiency.
- Automated and tracked **50+** experiments through reproducibility-first logging infrastructure built into the workflow pipeline.
- Re-engineered feature extraction pipelines for CIFAR-10 image classification, improving model accuracy by **8%**.

Projects

Nexora : AI-Powered Networking CRM Platform

[Link](#)

React 19 • Vite • Tailwind CSS • FastAPI • PostgreSQL • SQLAlchemy • Celery • Redis • Docker

- Architected role-based access control with rate limiting across **310+** REST endpoints, supporting dual JWT and Supabase authentication modes.
- 100% environment parity across **dev/prod**, achieved by containerizing the full stack (React, FastAPI, PostgreSQL, Redis) with Docker Compose.
- Automated regression coverage on all critical auth/API paths, achieved by building a CI/CD-ready pytest + Playwright pipeline.

Veridex: Hybrid Search and Grounded RAG Platform

[Link](#)

React • TypeScript • FastAPI • Qdrant • Gemini Embeddings • Docker • GitHub Actions

- Engineered a multi-format document retrieval platform for **TXT, Markdown, PDF, and DOCX**, using deterministic normalization, chunking, document identifiers, and metadata-filtered indexing in Qdrant.
- Combined **768-dimensional Gemini semantic embeddings** with sparse lexical retrieval through weighted Reciprocal Rank Fusion using **1.5 dense weight, 1.0 sparse weight, and k=60**, with branch-level ranking and score diagnostics.
- Developed a citation-validated grounded-answer pipeline with insufficient-context refusal, API-key authentication, readiness checks, and request observability; verified through **817 passing backend and frontend tests** and deployed on Vercel with Qdrant Cloud.

CodeForge : Real-Time Collaborative Code Editor

[Link](#)

React • Node.js • Express.js • MongoDB • Socket.IO • JWT • Judge0 API

- Built a MERN-stack collaborative code editor with real-time multi-user editing and room-based sessions via Socket.IO.
- **0% sandbox escapes** in testing, by integrating secure, isolated code execution via a protected `/api/execute` endpoint on **Judge0 CE**.
- Implemented JWT-authenticated access control for collaboration rooms, restricting session participation to authorized users.

Certifications

MongoDB Certified Associate Developer MongoDB, Inc.

[Certificate](#)

Schema design, aggregation pipelines, indexing, and CRUD on document-oriented databases.

Oracle Certified Professional: Java SE 17 Developer Oracle

[Certificate](#)

Validated proficiency in core Java, OOP design, exception handling, generics, and concurrency.